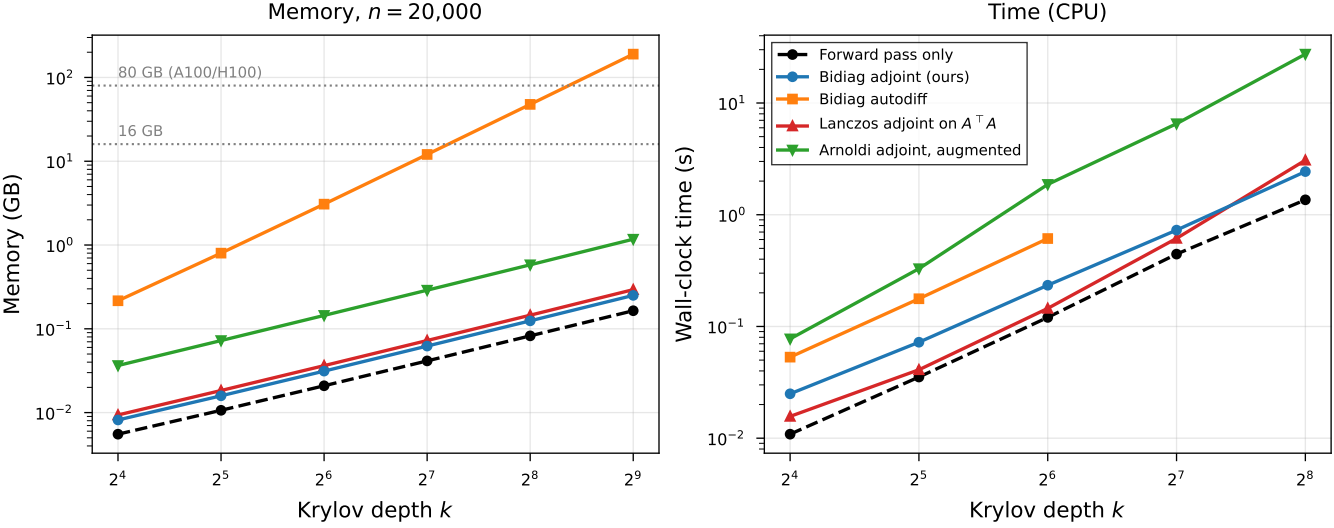
Memory, $n = 20{,}000$
80 GB (A100/H100)
16 GB
Memory (GB)
10^{2}
10^{1}
10^{0}
10^{-1}
10^{-2}
2^{4}
2^{5}
2^{6}
2^{7}
2^{8}
2^{9}
Krylov depth k
Time (CPU)
Forward pass only
Bidiag adjoint (ours)
Bidiag autodiff
Lanczos adjoint on $A^\top A$
Arnoldi adjoint, augmented
Wall-clock time (s)
10^{1}
10^{0}
10^{-1}
10^{-2}
2^{4}
2^{5}
2^{6}
2^{7}
2^{8}
Krylov depth k